



Effect of heat stress and herbal products supplementation on the immune response related gene expression patterns in Kenguri ewes

E. Binuni Rebez^{a,b}, C. Devaraj^a, J. Ninan^b, M.V. Silpa^b, S. Venkatesa Perumal^b,
Kumar Tej Nikhil^b, A. Sahoo^a, F.R. Dunshea^c, V. Sejian^{a,b,*}

^a Centre for Climate Resilient Animal Adaptation Studies, ICAR-National Institute of Animal Nutrition and Physiology, Adugodi, Bengaluru 560030, India

^b Rajiv Gandhi Institute of Veterinary Education and Research, Kurumbapet, Puducherry 605009, India

^c School of Agriculture, Food and Ecosystem Sciences, Faculty of Science, The University of Melbourne, Parkville, Melbourne, VIC 3010, Australia

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ABSTRACT

The study was conducted to establish the impact of heat stress as well as the herbal products supplementation on the expression patterns of immune response related genes in indigenous Kenguri ewes. A sixty days study was carried out in a climate-controlled environment by using 24 adult female Kenguri sheep. The study animals were randomly grouped into four with six animals each, viz., C (n = 6; Control; standard diet), HS (n = 6; Heat Stress; standard diet), CS (n = 6; Control Supplement; standard diet +herbal powder), and HSS (n = 6; Heat Stress Supplement; standard diet + herbal powder). The temperature humidity index (THI) recorded clearly indicated that the HS and HSS group ewes were exposed to severe heat stress while C and CS group ewes were exposed to thermoneutral condition. Further, heat stress increased plasma cortisol concentration while the herbal supplements reduced cortisol concentration (14.72 vs 11.16 ng/ml for HS and HSS, respectively, $P < 0.01$). Likewise, heat stress ($P < 0.05$) reduced the expression patterns of *Interleukin-2 (IL-2)*, *IL-18*, *Tumor necrosis factor alpha (TNF- α)*, *Interferon beta (IFN- β)*, *Interferon gamma (IFN- γ)*, *superoxide dismutase (SOD)* and *glutathione peroxidase (GPx)*. However, heat stress ($P < 0.05$) upregulated *IL-6* expression. Therefore, all these cytokines can serve as biomarkers for quantifying heat stress on the immune response in Kenguri ewes. Further, the results also established that the herbal supplementation reduced the heat stress induced increased cortisol concentration and *IL-6* levels (at day 60), in addition to recovering heat stress induced reduced levels of *IL-2*, *TNF- α* , *IFN- γ* and *SOD* to normal range. Thus, the herbal supplements used in this study brought beneficial effects in relieving the heat stress in Kenguri ewes.

1. Introduction

Climate change has been a significant threat to the livestock husbandry systems, globally. However, the ability of each species to adapt to the effects of elevated temperature and harsh environmental conditions differs widely. In this context, small ruminants are established to be well suited to warmer and harsh settings in contrast to large ruminants as they demonstrate extensive adaptive responses to curb the negative effects of heat stress (Joy et al., 2020). The small ruminants are known for their efficacy in delivering quality products by utilizing feeble resources with reference to feed, water and labour in addition to their versatile adaptability, climate resilient potential and small ecological-footprint (Joy et al., 2020; Sejian et al., 2021). Small

ruminants are thus featured as a critical livestock to ensure food security and economic benefits. Even if small ruminant species are considered to be rustic, their potential to endure elevated temperature, without compromising production and welfare has often been overestimated (Al-Dawood, 2015). The detrimental effects of heat stress on sheep production includes decreased growth performance, reproduction, milk production as well as immune compromise that predisposes animals to a wide range of diseases and also leads to mortality (Al-Dawood, 2017). In order to prevent economic losses associated with production losses, it is vital to maintain animal health as a measure to cope well with the stressors associated with hot environmental conditions.

The elevated temperature and humidity causing significant endocrine disruptions in order to facilitate heat loss in animals have a

* Corresponding author at: Centre for Climate Resilient Animal Adaptation Studies, ICAR-National Institute of Animal Nutrition and Physiology, Adugodi, Bengaluru 560030, India

E-mail address: drsejian@gmail.com (V. Sejian).

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negative influence on the immune functions. The stressors contributing to elevated cortisol levels have been associated with reducing the immune functions (Rhind et al., 1999). Primarily, the higher cortisol concentration during acute stress suppresses cell proliferation and its post-stressor levels have a negative impact on immune responses, with a bidirectional effect based on the duration of stress (Ciliberti et al., 2017). The co-occurrence of cortisol and prolactin exerts adverse effects on immune-related genes under chronic heat stress (Collier et al., 2008). Further, cortisol has immune modulatory role in heat stressed sheep reflected by altered concentrations of interleukins (Caroprese et al., 2018). Thus, in heat stressed animals, the cortisol concentrations have been demonstrated to have a significant influence on the immune response. Further, heat stress in sheep causes significant alterations in cytokine population and immunoglobulin indicating that severe heat stress leads to detrimental impact on the immune function (Shi et al., 2020). The relation between heat stress and immune response in livestock rely on complex mechanisms including altered body temperature, behavioral and endocrine adaptive responses, circulatory adjustments and oxidative stress (Lacetera, 2012; Shi et al., 2020). However, the mechanism underpinning impaired immune system of sheep raised under high temperature conditions remains undefined. There is a considerable variation in immune potential between crossbreds and native breeds. The superiority of indigenous animals over cross breeds has been highlighted by their immune competency during heat stress conditions (Bhanuprakash et al., 2016).

Further, in addition to the innate potential of indigenous sheep to combat heat stress conditions, optimal nutrition can enhance their adaptive response and ensure optimal energy source for coping with the stressors (Sejian et al., 2014). There are established evidences about nutritional interventions in ameliorating the detrimental effects of heat stress in sheep and among the various nutritional supplements herbal remedies have been found to be effective (Rebez et al., 2024). The herbs and the botanicals have huge potential in stimulating the immune system, improving growth performance and antioxidant status of heat stressed sheep (Hashemzadeh et al., 2022; Saleh et al., 2023). There are established evidence about the effect of herbal feed additives like amla (Lakhani et al., 2017), tulsi (Srivastava and Singh, 2021), noni (Zhang et al., 2022), ashwagandha (Tiwari and Sahni, 2012) and bamboo (Li et al., 2021) in stress amelioration and immune defense in different livestock species. However, scientific evaluation and standardization of herbal products in ameliorating heat stress across various livestock species have remained unaddressed. In addition, there is a paucity of evidence regarding the immune regulation and anti-oxidant potential of native sheep breeds under high temperature conditions. While previous studies have examined the heat stress protection effects of individual herbs, there remains a lack of research on the combination of herbal supplements, which this study aims to address. The combination of herbal feed additives may be highly beneficial rather than individual herbs in imparting multiple benefits to the heat stressed animals. With this background, the objective of the current study was to evaluate the efficacy of herbal products supplement on the immune regulation of heat stressed indigenous Kenguri sheep, based on cortisol concentrations and expression pattern of immune related genes including *Interleukin-2* (IL-2), *IL-6*, *IL-18*, *Tumour Necrosis Factor-alpha* (TNF- α), *Interferon-beta* (IFN- β), *Interferon gamma* (IFN- γ), *superoxide dismutase* (SOD), and *glutathione peroxidase* (GPx).

2. Materials and methods

2.1. Study location and climate

The study was carried out at the Centre for Climate Resilient Animal Adaptation Studies (CCRAAS), ICAR-National Institute of Animal Nutrition and Physiology (ICAR-NIANP), Bangalore, India. The site of the study is located at longitude 12°57'04.3"N, latitude 77°36'25.3"E and an altitude of 920 m above mean sea level in the southern Deccan

plateau of the country. The annual average ambient temperature ranges between 15°C and 36°C. The average relative humidity (RH) ranges from 20 % to 85 % annually. This geographic area receives 200–970 mm of rain every year and is unevenly distributed across the year. The trial was conducted between the months of June and August. The annual minimum and maximum temperatures fluctuates between 15 and 22°C and 27–34°C, respectively. The yearly RH varied from 28 to 59 percent during the study period, indicating hot, semi-arid conditions.

2.2. Animals

Kenguri or Tenguri breed of sheep is popularly reared for mutton production in Karnataka state that features a tropical climate. The breed which is named after its dark red color body coat has a characteristic medium sized droopy ear with a short tail and the average body weight of adult males ranged between 32 and 72 kg and that of the adult females ranged between 23.5 and 57.5 kg (NBAGR, 2020). In the present study, 24 young adult female Kenguri sheep in active growth phase (1–2 year old) with an approximate body weight ranging between 15–23 kg were used. The animals were maintained in clean, well-ventilated shed with asbestos roofing and with access to good quality feed and water. The prophylactic interventions like vaccination and deworming against common sheep infectious diseases and parasitic infestations were carried out in accordance to the standard health management plan.

2.3. Study design

A sixty days study was carried out in a climate-controlled environment by using 24 adult female Kenguri sheep. The study animals were randomly grouped into four with six animals each, viz., C (n = 6; Control; standard diet), HS (n = 6; Heat Stress; standard diet), CS (n = 6; Control Supplement; standard diet +herbal powder), and HSS (n = 6; Heat Stress Supplement; standard diet + herbal powder). The study animals were stall-fed with a diet composed of 60 % roughage (Hybrid Napier - *Pennisetum purpureum*) and 40 % concentrate mixture (maize, 36 kg/100 kg; wheat bran, 37 kg/100 kg; soybean meal, 25 kg/100 kg; mineral mixture, 1.5 kg/100 kg and common salt, 0.5 kg/100 kg with Crude Protein (CP) = 23.15 % and Total Digestible Nutrients (TDN) = 71 %. Table 1 details the feed ingredients and chemical composition

Table 1
The feed ingredients and the chemical composition of the roughage and concentrate ration administered to the trial animals.

Attribute	Concentrate mixture (kg/100 kg)	Napier hay (<i>Pennisetum purpureum</i>)
Ingredients	36	-
Maize		
Wheat bran	37	-
Soybean meal	25	-
Mineral mixture	1.5	-
Salt	0.5	-
Chemical composition (%)	90.83	92.32
Dry matter		
Organic matter	94.02	88.59
Crude protein	23.15	6.84
Ether extract	3.77	2.72
Crude Fibre	6.12	32.58
Total ash	5.98	11.41
Fibre fractions (%)	29.13	71.51
Neutral detergent fibre		
Acid detergent fibre	8.97	43.94
Nutritive value	71	53.20
Total digestible nutrients (%)		
Digestible energy (kJ/kg)	14.24	9.78
Metabolizable energy (kJ/kg)	11.78	7.98

provided for the study animals. Table 2 details the composition of the herbal supplement used in the experiment. The herbal supplement composed of a mixture of *Ocimum sanctum* (Tulsi), *Embllica officinalis* (Amla), *Morinda citrifolia* (Noni), *Withania somnifera* (Ashwagandha) and *Phyllostachys edulis* (Bamboo). The herbal products supplement used in the present study was supplemented to CS and HSS group animals in dry powder form at a dose of 0.8 g/Kg BW/Day along with the concentrate ration. The herbal supplement was weighed based on individual animal body weight every week and mixed in the respective animal concentrate mixture and provided daily. In subsequent weeks based on that particular week body weight the herbal supplement was weighed and mixed in concentrate offered for individual animals. This procedure was followed throughout the study. The animals in C and CS group were placed in thermo-neutral chamber and exposed to 24°C, from 10:00 h to 16:00 h for each day throughout the study duration of sixty day in order to maintain comfort conditions. The animals in HS and HSS group were maintained in the heating chamber and were exposed to heat stress under simulated heat stress model described in Table 3. During the trial, the animals in the heating chamber and thermo-neutral zone chamber were exposed to respective temperatures for six hours each day, from 10:00 h to 16:00 h.

Both the thermal comfort zone climate chamber and heating climate chamber are designed to provide the appropriate environmental conditions required for any environmental stress-related studies. It is an established fact that the thermal comfort zone for sheep ranges between 20 and 28°C (de Sousa Silva et al., 2024). Therefore, the thermal comfort zone chamber was maintained at 24°C throughout the study period for six hours duration daily between 10:00 h to 16:00 h to provide the ideal thermal comfort zone to the C and CS group ewes. In the heating climate chamber, the HS and HSS group ewes were exposed to a simulated heat stress model as described in Table 3. The simulated heat stress model described in Table 3 has been designed using baseline meteorological data during summer season on temperature and relative humidity for each hour during the six hours experimental period daily using a programmable thermostat to progressively increase the respective hour temperature and humidity. Thus, the weather variables were controlled only during these six hours and during the rest of the time (from 16:00 h to 10:00 h), the animals in both the chambers were maintained without simulation and the average THI during this remaining period was 74.38 and 74.45 respectively for TNZ and heating chamber. The model was developed with the intention of mimicking the natural environmental stress condition in the climate chamber. This experimental set up was based on replicating field conditions wherein the animals during grazing are exposed to heat stress for a span of six hours from 10:00 h to 16:00 h. Moreover, from an animal ethical point of view, the simulated heat stress model was more practical and less stressful than the constant heat stress models.

The feeding and watering activity was done in individual animal enclosures throughout the study period. The blood sampling for hormonal analysis was done at fortnightly intervals. After the end of the experiment, blood samples were collected for peripheral blood mononuclear cell (PBMC) isolation in order to carry out gene expression

studies. The experimental procedures, including subsection of experimental animals to heat stress were conducted after the approval from the institute animal ethics committee letter number V-11011(13)/14/2021-CPCSEA-DADF.

2.4. Blood sampling for endocrine analysis

The blood collection for plasma cortisol estimation was carried out at fortnightly intervals. The blood was collected (4 ml) using heparin coated vacutainers from the external jugular vein of the animals in all groups. The blood was drawn from the animals in all the four groups simultaneously at 11:00 h. The blood samples were immediately centrifuged at 3500 rpm for 20 minutes and plasma was harvested and stored at -20°C until further analysis.

The concentration of the target hormone in the sample was estimated using the Enzyme Linked Immuno Sorbent Assay (ELISA) method (Puregene kit, Genetix, New Delhi). Before commencing the reaction procedure, the reagents, standard solution, and samples were thawed and prepared according to the guidelines. Further, the reagents used in the reaction were thoroughly mixed without foaming. Following the start of the procedure, all of the steps were carried out without any interruptions. To start with, 50 µL of standard solution was added to the standard well. Further, 40 µL of the sample and 10 µL of the anti-GH antibody were added to the sample wells. After that, 50 µL of streptavidin-HRP was added to the standard and sample wells (the blank control well was ignored). After gently mixing the plates, it was sealed and incubated at 37° C for sixty minutes. After removing the sealer, the plates were washed at least five to six times with 400 µL/well/wash of the prepared wash buffer. The plates were then blotted by striking them firmly against the absorbent paper to get rid of any residual droplets (by hand six times). Following this, at predetermined intervals, 50 µL of substrate solution A and an equivalent volume of substrate solution B was pipetted into each well. After applying a sealer, the plates were then placed in the dark at 37° C for ten minutes. Then, 50 µL of stop solution was added to all wells and the change in the color from blue to yellow was evidenced and the optical density (OD) value was measured using a microplate reader set to 450 nm within ten minutes of adding the stop solution. Further, using immunoassay software, the OD of the samples were compared to the standard curve and studied.

2.5. Blood sampling for PBMC isolation

At the end of the trial, blood collection for PBMC isolation was carried out at 11:00 h simultaneously from all the animals. Approximately five ml of blood samples were collected from external jugular vein in EDTA coated vacutainers and were immediately stored in ice. Following blood collection, the samples were processed for PBMC isolation. To the blood samples ice-cold 1 × RBC lysis buffer was added and incubated at room temperature for 10 min followed by centrifugation at 2500 rpm at 4°C for 5 min. The total RNA was isolated from the PMBCs using the GeneJET Whole Blood RNA Purification Mini Kit (Thermo Scientific, Lithuania) in accordance with the manufacturer's instructions with a few minor adjustments as mentioned: the pellet was re-suspended in 600 µL of lysis buffer containing 20 µL of 2- mercaptoethanol, mixed well by vortexing. Further, 600 µL of buffer containing 590 µL of Tris EDTA and 10 µL proteinase K was added, vortexed and incubated for one hour. Then, 450 µL of ethanol (96– 100 %) was added and mixed well by vortexing. The lysate was transferred to the purification column and centrifuged at 12,000 rpm for 2 min, and the flow-through was discarded. After reassembling the column and collection tube, the remaining lysate was transferred and centrifuged similarly as before and the collection tube along with the flow-through was discarded. Following this, 700 µL of wash buffer 1 was added to the purification column and centrifuged for 2 min at 12,000 rpm, followed by discarding the flow-through solution. Then, 600 µL of wash buffer 2 was added to the purification column and centrifuged at the rate of 12,000 rpm for

Table 2

The Composition of Herbal Supplement fed to the animals in the study.

Nutrient	Composition (100 g)
<i>Ocimum sanctum</i>	30 g
<i>Embllica officinalis</i>	30 g
<i>Morinda citrifolia</i>	20 g
<i>Withania somnifera</i>	10 g
<i>Phyllostachys edulis</i>	10 g
Dose rate	0.8 g/Kg BW/Day

The herbal supplement in dry powder form was supplemented to CS and HSS group animals at a dose of 0.8 g/Kg BW/Day along with the concentrate ration

Table 3

The Simulated Heat Stress Model for the trial.

Time	10.00 h	11.00 h	12.00 h	13.00 h	14.00 h	15.00 h	16.00 h
Temperature (°C)	36.0	37.5	38.5	38.8	40.0	38.3	37.0
Relative Humidity (%)	49.0	44.0	41.0	40.0	39.0	42.0	45.0

The values are average at that particular hour for the entire summer duration (April-June) for thirty years duration

2 min, followed by discarding the flow-through solution. Likewise, 500 µL of wash buffer 2 was added again to the purification column and centrifuged, and the flow through was discarded. The empty purification column was re-spun at 12,000 rpm for 2 min. The nuclease-free water (NFW) of 20 µL was added at the centre of the column and centrifuged for 5 min at the rate of 12,000 rpm. Fresh nuclease-free micro-centrifuge tubes were used to collect the eluted RNA, which was processed immediately following collection.

RevertAid First Strand cDNA Synthesis Kit (Thermo Scientific, Lithuania) was used to reverse transcribe the total RNA into complementary DNA (cDNA) for use in real-time quantitative polymerase chain reaction (RT-qPCR). This was followed as per the manufacturer's protocol as follows: 4 µL of 5 × Reaction Mix, 1 µL RevertAid M-MuLV Reverse Transcriptase (RT) (200 U/µL), 5 µg of template RNA, 15–20 pmol of primer, 1 µL of RiboLock RNase Inhibitor (20 U/µL), 2 µL of 10 mM dNTP Mix and NFW was added to make the volume to 20 µL. Following this, the contents were mixed gently, and subjected to reverse transcribing PCR (5 min at 25°C, followed by 42°C for 1 h, and the enzymes was inactivated by heating at 70°C for 5 min). The product of the first-strand cDNA synthesis was diluted to a final concentration of 25 ng/µL with NFW and 2 µL of diluted cDNA was used for each reaction in qPCR. An online NCBI primer design software was used to create a gene-specific primer, and its specificity was verified using BLAST and Primer3 (<http://www.ncbi.nlm.nih.gov/tools/primer-blast/>). Table 4 describes the primer sequences of the housekeeping and target genes.

The relative expression of the targeted genes was investigated using SYBR green chemistry (Maxima SYBR green qPCR master mix, Fermentas, USA). The 20-µL reaction was carried out in triplicate using 50 ng of template and 0.5 µM primer concentrations. The reaction condition was enzyme activation at 95 °C for 10 min and amplification cycle (40 cycles; initial denaturation at 95 °C for 15 s, annealing at 60 °C for 30 s, and extension at 72 °C for 30 s). By examining the melt curve analysis towards the end of the reaction, non-specific amplification was ruled out. As an internal control, the housekeeping gene *GAPDH* (glyceraldehyde 3-phosphate dehydrogenase) was utilised to compare the

relative expression of target genes. By using the $2^{\Delta\Delta CT}$ method, the relative expression patterns of the target genes were analysed.

2.6. Variables studied

The targeted weather variables in the study were as follows: maximum temperature, minimum temperature and relative humidity which were recorded using thermo-hygrometer and dry bulb temperature, wet bulb temperature which were recorded using dry bulb and wet bulb thermometer, respectively. The plasma cortisol was estimated by using Puregene- enzyme-linked immunoassay (ELISA) Kit and a micro-plate reader. The targeted genes in the current study were Interleukin-2 (IL-2), IL-6, IL-18, Tumor necrosis factor alpha (TNF-α), Interferon beta (IFN-β), Interferon gamma (IFN-γ), superoxide dismutase (SOD) and glutathione peroxidase (GPX)

2.7. Data analysis

The endocrine and relative gene expression data were analyzed by using SAS University Edition (SAS/STAT®, SAS Institute Inc., Cary, NC, USA) applying the MIXED procedure to infer fixed, random, interaction and repeated measurement (for plasma cortisol) effects via mixed model equations. The statistical model adopted provided the output for the interaction effects along with the fixed factors. The statistical model adopted was:

$$Y_{ijklmnop} = \mu + F_i + T_j + D_k + F \times T_1 + F \times D_m + T \times D_n + F \times T \times D_o + A_p + e_{ijklmnop}$$

where $Y_{ijklmnop}$ was the observation for the plasma cortisol and relative gene expression traits from the $ijklmnop^{th}$ sheep, μ was the overall mean effect, F_i was the fixed effect for the i^{th} diet (i = standard diet, standard diet + herbal supplementation), T_j was the fixed effect for the j^{th} temperature (j = thermoneutral, heat stress), D_k was the fixed effect for the k^{th} day (applied only while analyzing plasma cortisol; k = day0, day15, day30, day45, day60), $F \times T_1$ was the l^{th} interaction

Table 4

The Primers used for studying the relative expression of the targeted genes.

Gene ID		Primer Sequence (5'-3')	Product size (bp)	Accession no.
GAPDH	F	GGTGATGCTGGTGCTGAGTAG	147	XM_005680968.3
	R	ACGATGTCCACTTTGCCAGT		
IL-2	F	CCTGAGCAGGATGGAGAAATTACA TCCAGAACATGCCGCAGAG	141	NM_008366.3
	R			
IL-6	F	TCCAGTTGCCTTCTTGCGAC GTGTAATTAAGCCTCCGACTTG	140	NM_031168.1
	R			
IL-18	F	CATACGAAATTTGAACGACCA	154	XM_012095263.3
	R	TACAGCCAGACCTCTAGTGA		
IFNβ	F	GGAGCCAATCCGGAAGGAAA	108	JX458084.1
	R	GACTCGAGGTACTGCACGAG		
IFNγ	F	AGGTCATTCAAAGGAGCATGG CTCAGAGCTGCCGTTCAGAA	72	NM_001009803.1
	R			
TNFα	F	CCATGAGGGCAATTGGCATACTCTCTCAAGCCTCAAGTAACAAGC	103	EU276079
	R			
SOD	F	GGAGACCTGGGCAATGTGAA	182	NM_001100919.3
	R	CCTCCAGCGTTTCCAGTCTT		(Su,2016)
GPX	F	TGGTCGTACTCGGCTTCC	163	XM_004018462.1
	R	AGCGGATGCGCCTTCTCG		(Jiang,2015)

bp Base pair, F Forward, R Reverse, GAPDH glyceraldehyde-3-phosphate dehydrogenase, IL-2- interleukins 2; IL-6- interleukins 6, IL-18 interleukins 18, TNF-α -Tumor necrosis factor alpha, IFNβ-Interferon beta, IFN-γ -interferon gamma, GPX- glutathione peroxidase,SOD- Superoxide dismutase

between feed and temperature, $F \times D_m$ was the m^{th} interaction between feed and day (applied while analyzing plasma cortisol), $T \times D_n$ was the n^{th} interaction between temperature and day (applied only while analyzing plasma cortisol), $F \times T \times D_o$ was the o^{th} interaction between feed, temperature and day (applied while analyzing plasma cortisol), A_p was the random animal effect for the p^{th} animal and $e_{ijklmnop}$ was the random residual effect. The least square means were estimated for all the fixed and interaction effects.

3. Results

3.1. Temperature humidity index (THI)

The average THI recorded in the thermoneutral zone chamber and heating chamber were 68.93 and 93.34, respectively (Table 5). The THI was recorded as per formula described by McDowell, (1972). As per this formula, any THI value above 78 is considered to be an extremely severe heat stress condition while any value below 72 was considered to provide comfortable condition for the animals. Thus the obtained average THI value of 93.34 recorded in heating chamber ensured that the HS and HSS groups ewes were subjected to extremely severe heat stress. Further, the THI of 68.93 obtained in the thermoneutral chamber clearly indicated that the C and CS groups ewes were subjected to comfort condition. The average THI from 16.00 h to 10.00 h was 74.38 and 74.45, respectively in TNZ and heating chamber with negligible difference between the chambers.

3.2. The effect of heat stress and herbal product supplementation on plasma cortisol in Kenguri sheep

The plasma cortisol concentration was significantly influenced by temperature ($P < 0.05$), diet ($P < 0.01$) and day ($P < 0.001$). Further, temperature $\times$ diet, and diet $\times$ day interactions highly ($P < 0.001$) influenced the plasma cortisol concentration on all fortnightly intervals (Table 6). The day 60 plasma cortisol profile revealed significant impact of herbal supplement on reducing the heat stress induced alteration in plasma cortisol. An increase in plasma cortisol was observed on day 60 in heat stress sheep fed with standard diet (11.5 ± 0.57 vs 14.2 ± 0.57 ng/ml for C vs HS, respectively). Whereas the same for sheep with herbal supplementation were comparable (9.09 ± 0.57 vs 9.16 ± 0.57 ng/ml for CS vs HSS, respectively).

4. The effect of heat stress and herbal product supplementation on the expression pattern of targeted genes in Kenguri sheep

4.1. Expression patterns of *IL-2*, *TNF- α* , *IFN- γ* and *SOD*

The effect of heat stress and herbal product supplementation on *IL-2*, *TNF- α* , *IFN- γ* and *SOD* expression patterns in Kenguri sheep were described in Fig. 1a, d, f, and g, respectively. The expression patterns of all these genes were similar in nature across all four groups. Among the different groups, the significantly ($P < 0.01$) lower expression of *IL-2*, *TNF- α* , *IFN- γ* and *SOD* were recorded in HS group with no significant difference in expression pattern between C, CS and HSS groups. The expression of *IL-2*, *TNF- α* , *IFN- γ* and *SOD* were significantly reduced by heat stress (1.06 vs 0.44 ; 1.14 vs 0.40 ; 1.12 vs 0.40 ; 1.20 vs 0.14 , respectively; $P < 0.01$). However, the herbal supplements significantly

increased ($P < 0.05$) this heat stress induced reduced expression of all these four genes (0.44 vs 0.80 ; 0.4 vs 0.83 ; 0.4 vs 0.83 and 0.14 vs 0.56 , for *IL-2*, *TNF- α* , *IFN- γ* and *SOD*, respectively). Further, the heat treatment and diet interaction did not influence *IL-2*, *TNF- α* , *IFN- γ* and *SOD* expression patterns.

4.2. Expression pattern of *IL-6*

The effect of heat stress and herbal product supplementation on *IL-6* expression pattern in Kenguri sheep was described in Fig. 1b. Among the different groups, the significantly higher expression of *IL-6* was recorded in HS group as compared to other three groups. The heat stress significantly ($P < 0.001$) increased the expression pattern of *IL-6* (1.02 ± 0.29 vs 2.80 ± 0.29), but this significant increase in expression of *IL-6* was significantly lowered by supplementation with herbs (2.80 ± 0.29 vs 0.39 ± 0.29 ; $P < 0.001$). Further, there was a significant ($P < 0.01$) interaction between heat treatment and diet such that the herbal supplementation lowered the *IL-6* expression during heat stress (2.80 ± 0.29 vs 0.39 ± 0.29) but not under thermoneutral condition (1.02 ± 0.29 vs 1.10 ± 0.29).

4.3. Expression patterns of *IL-18*, *IFN- β* , and *GPX*

The effect of heat stress and herbal product supplementation on *IL-18*, *IFN- β* , and *GPX* expression patterns in Kenguri sheep were described in Fig. 1c, e and h. Among the different groups, the significantly ($P < 0.01$) lower expression of *IL-18*, *IFN- β* , and *GPX* were recorded in HS and HSS groups as compared to C and CS groups. The expression of *IL-18*, *IFN- β* , and *GPX* were significantly reduced by heat stress (1.08 vs 0.50 ; 1.20 vs 0.47 ; and 1.09 vs 0.27 , respectively; $P < 0.01$). However, this heat stress induced reduction in expression patterns of *IL-18*, *IFN- β* , and *GPX* were not recovered by herbal supplements (0.50 vs 0.75 ; 0.47 vs 0.58 ; and 0.27 vs 0.63 , for *IL-18*, *IFN- β* , and *GPX*, respectively). Further, the heat treatment and diet interaction did not influence *IL-18*, *IFN- β* , and *GPX* expression patterns.

5. Discussion

The central nervous and immune systems are connected through the activation of the autonomic nervous system and the hypothalamic-pituitary-adrenal (HPA) axis by the secretion of numerous molecules such as neuropeptides and neurohormones, which are responsible for changes in cellular and humoral immune responses (Sevi et al., 2001; Caroprese et al., 2010). During stress, various endocrine responses are involved to improve the fitness of the individual. The front-line hormones to overcome stressful situations are the glucocorticoids and catecholamines (Sejian and Srivastava, 2010). Chronic or long-term stress can suppress immunity by decreasing immune cell numbers and function and/or increasing active immunosuppressive mechanisms. Chronic stress can also dysregulate immune function by promoting proinflammatory and type-2 cytokine-driven responses (Dhabhar, 2009). Thus, this study elucidates the relationship between HPA product cortisol and associated immune system related cytokines gene expression patterns.

The heat stress significantly increased the plasma cortisol concentration in the present study. It is a well-established fact that heat stress in sheep leads to increased blood cortisol concentration (Sevi et al., 2002;

Table 5

Average weather variables recorded during the entire study period.

	Time of recording	DBT (°C)	WBT (°C)	MaxT (°C)	MinT (°C)	AT (°C)	RH (%)	THI
Thermo-neutral zone Chamber	Afternoon (14:00 h)	20.15 $\pm$ 0.05	19.19 $\pm$ 0.07	28.49 $\pm$ 0.62	23.98 $\pm$ 0.21	26.23 $\pm$ 0.32	77.42 $\pm$ 1.76	68.93 $\pm$ 0.07
Heating Chamber	Afternoon (14:00 h)	40.44 $\pm$ 0.18	32.81 $\pm$ 0.31	36.92 $\pm$ 0.42	27.92 $\pm$ 0.53	32.57 $\pm$ 0.33	69.45 $\pm$ 1.41	93.34 $\pm$ 0.34

DBT- dry bulb temperature; WBT- wet bulb temperature; THI- temperature-humidity index; MaxT- maximum Temperature; MinT- minimum temperature; RH- relative humidity; AT- Ambient Temperature; RH- relative humidity

Table 6

Effect of heat stress and herbal products supplementation on plasma cortisol level in Kenguri ewes.

Variable	Temperature (T)	Diet	Day					Significance
			0	15	30	45	60	
Cortisol (ng/ml)	Thermoneutral	Control	13.4 ± 0.99 ^{aA}	12.8 ± 1.12 ^{aB}	12.1 ± 0.65 ^{aB}	11.7 ± 0.84 ^{aB}	11.5 ± 0.57 ^{aB}	Temperature.Diet**, Diet. Day**
		Herb	13.6 ± 0.99 ^{aA}	12.4 ± 1.12 ^{aB}	12.5 ± 0.65 ^{aB}	9.70 ± 0.84 ^{bb}	9.09 ± 0.57 ^{bc}	
	Heat Stress	Control	13.5 ± 0.99 ^{aA}	15.4 ± 1.12 ^{aA}	15.6 ± 0.65 ^{aA}	14.9 ± 0.84 ^{aA}	14.2 ± 0.57 ^{aA}	Day*** Diet** Temperature*
		Herb	13.6 ± 0.99 ^{aA}	12.1 ± 1.12 ^{abB}	11.1 ± 0.65 ^{bcB}	9.84 ± 0.84 ^{bcB}	9.16 ± 0.57 ^{cc}	
		Day effect	13.5 ± 0.49 ^a	13.2 ± 0.56 ^a	12.9 ± 0.32 ^a	11.5 ± 0.42 ^b	11.0 ± 0.29 ^b	
		Diet effect	13.5 vs 11.3					
		Temperature effect	11.9 vs 12.9					

Significant differences between temperature x diet within each day (column) are denoted as A,B,C. Significant differences between diet x day within each temperature (row) are denoted as a,b,c.

* Significant at $P < 0.05$; NS-Non Significant

** Significant at $P < 0.01$;

*** Significant at $P < 0.001$;

(Slimen et al., 2019). The cortisol hormone plays a very important role in supporting and controlling adaptive functions through energy production and thermoregulation in sheep (Slimen et al., 2019). Further, glucocorticoids regulates the catabolic response to heat stress through modulation of carbohydrate, protein and lipid metabolism to ensure continuous energy supply for sustaining life supporting activities in heat stressed animals (Hough et al., 2013). Thus, activation of hypothalamo-pituitary axis (HPA) forms a crucial step for relieving heat stress in animals. This increased HPA-axis activation increases the plasma cortisol secretion probably to increase glucose availability and to meet out the enhanced energy demand for thermoregulation (Matteri et al., 2000). Similar to our finding in Kenguri sheep, Rathwa et al. (2017) reported increased cortisol concentration in heat stressed indigenous sheep. Further, Karthik et al. (2021) also reported increased cortisol concentration in heat stressed Nellore sheep and attributed this increase in cortisol level to aid in vasodilation to facilitate heat loss. In the current study, the highest cortisol concentration was recorded on day 15 and lowest concentration was recorded on day 60. Similar to our finding, Wojtas et al. (2014) reported increased cortisol concentration during acute heat stress exposure, however, the level decreased gradually as the experimental days progressed in the climate chamber. These authors attributed decrease in cortisol concentration levels during chronic heat stress exposure as a result of animal's adaptation to acute heat stress conditions. This heat stress induced increase in cortisol concentration was significantly reduced by herbal supplementation during heat stress exposure in Kenguri sheep. This reflects the heat stress protection effects of herbal supplement in the current study. There were no reports available in the existing literatures to understand the governing mechanism by which these herbal products reduce the heat stress induced increase in cortisol concentration. Thus, these protective effects could be due to the combination of antioxidant, immune-protective, anti-inflammatory and cardio-protective effects of these supplements.

The heat stress significantly reduced the *IL-2* expression pattern in this study. The *IL-2* and *IL-6* could possibly play a key role to elicit the immune response to ameliorate the thermal insults during long-term heat stress acclimation (Rawash et al., 2022). Also *IL-2* was produced by the T-cell line plays an important role in immune-modulation (Madhusoodan et al., 2020) as well as activates the hypothalamo-pituitary-adrenal axis to adapt to the environmental stress (Wang et al., 2015). Similar to our findings, Shi et al. (2020) reported reduced *IL-2* expression in Dorper × Mongolian crossbred male lambs during heat stress exposure. These authors attributed the reduction of this cytokine to the increase in cortisol level of lambs under heat stress (Nazifi et al., 2003), which might cause a shift from a Th1 immune-driven response to a Th2 response (Wiegiers et al., 2005), suppressing the secretion and effect of *IL-2*. However, there are also contrasting reports of increased *IL-2* expression recorded during heat stress exposure in ewes (Rawash et al., 2022; Ashrafi et al., 2024). Further, the herbal supplements significantly

upregulated the heat stress induced reduction in *IL-2* expression reflecting the heat stress protection effects of herbal feed additives used in this study. In addition, the similar *IL-2* expression between C and HSS groups indicates that the herbal supplements brought back the level to normal range indicating the beneficial role of herbal supplements during heat stress exposure.

Heat stress increases plasma glucocorticoid levels in a process that negatively affects the synthesis and release of certain cytokines, especially *TNF-α* (Bagath et al., 2019). Similar to this finding, heat stress increased the plasma cortisol concentration and reduced the expression pattern of *TNF-α* in Kenguri ewes in this study. However, Shi et al. (2020) reported increased expression of *TNF-α* in Dorper × Mongolian crossbred male lambs during chronic heat stress exposure. Similar heat stress induced increased expression of *TNF-α* was also established in crossbred sheep (Chauhan et al., 2014). In a study in Holando-Argentino cows it was demonstrated that the *TNF-α* expression diminished drastically during summer and it was concluded that animals that are not physiologically prepared to resist high temperatures might have a less efficient immune response, reinforcing the need to develop new strategies to improve animal welfare (Lendez et al., 2021). The herbal supplements significantly upregulated the heat stress induced reduced *TNF-α* expression. This indicates the immune protective effects of the herbal supplements used in this study.

Likewise heat stress induced decreased expression pattern of *IFN-γ* was established in Kenguri ewes. Hyperthermia can downregulate Th1 cytokines like *IFN-γ* in favor of the secretion of Th2 cytokines, thus suppressing cell-mediated immunity (Caroprese et al., 2014). Further, heat stress-associated increased cortisol production can inhibit the expression of *IFN-γ* (Liu et al., 2012). Similar to this finding, the increased plasma cortisol, especially at day 60 in the current study could have played a role in downregulating *IFN-γ* expression. The decreased *IFN-γ* expression could reflect the compromised Th1 type immune response in Kenguri ewes (Chu and Song, 2013). Further, the herbal supplements used in this study significantly restored the heat stress induced reduced *IFN-γ* expression. This is evident from the significant differences in *IFN-γ* expression pattern between HS and HSS groups and non-significant difference between C and HSS groups. This could again reflect the immuno protective effects of the herbal diet used during heat stress exposure in Kenguri ewes. Thus, it could be inferred that the herbal supplements could restore the Th1 type immune response in Kenguri ewes during heat stress exposure in par with the thermoneutral control (C) group ewes.

In contrast to all immune traits in this study, heat stress significantly increased the expression pattern of *IL-6*. The *IL-6* is considered pro-inflammatory cytokines produced during stressful periods and regulate the acute phase response to stress (Kushner, 1993). Further, Al-Zghoul et al. (2019) reported *IL-6* to have a role in tissue protection mechanisms during heat stress exposure. This suggests that increased levels of *IL-6*

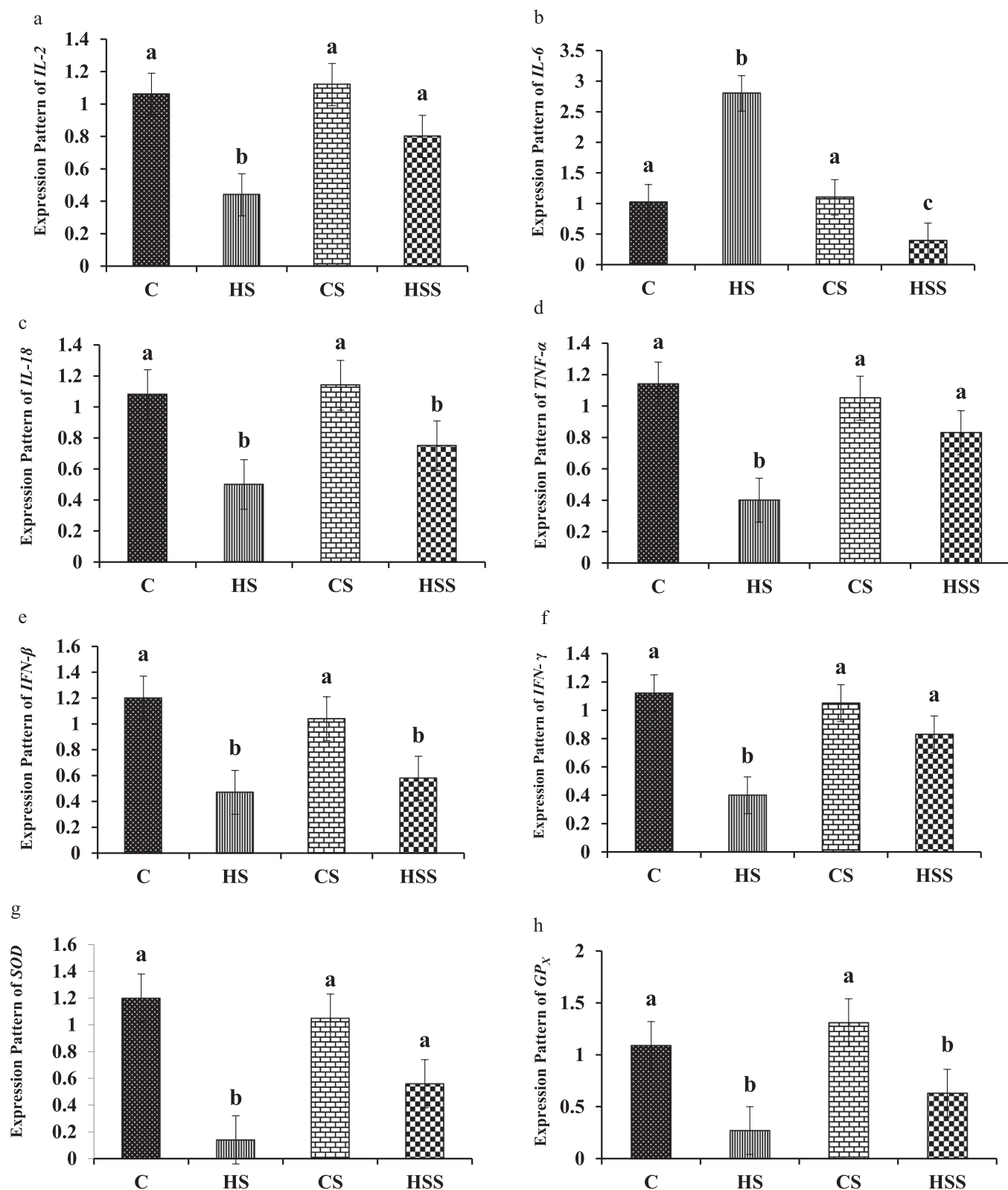


Fig. 1. Effect of heat stress and herbal product supplementation on different immunological genes mRNA expression pattern (Fold Change) in Kenguri sheep C (Control), HS (Heat Stress), CS (Control Supplement) and HSS (Heat Stress Supplement). The values bearing similar superscript do not differ with each other and the values bearing different superscript differ at $P < 0.05$. The herbal supplement composed of a mixture of *Ocimum sanctum* (Tulsi), *Embolica officinalis* (Amla), *Morinda citrifolia* (Noni), *Withania somnifera* (Ashwagandha) and *Phyllostachys edulis* (Bamboo). The non-supplemented group (C) is the control.

may regulate the tissue protection process during heat exposure. However in contrast to our findings, Caroprese et al. (2018) reported that heat stress significantly reduced the pro-inflammatory cytokine *IL-6* in sheep. Further, heat stress increases the secretion of glucocorticoids which acts as inhibitor of the pro-inflammatory cytokines such as *TNF- α* , *IL-6*, *IL-8* initiating the innate immune responses by the inhibition of the p38 MAPK pathway which maintains the stability of the immune system in animals. The day 60 plasma cortisol profile can support this statement wherein sheep fed with the standard diet (no herbal supplementation) and subjected to heat stress (HS) had a significantly higher plasma cortisol level (14.2 ng/ml) when compared to its control (C=11.5). While no significant difference in the plasma cortisol profile was observed in herbal supplemented sheep (CS=9.09 ng/ml; HSS=9.16 ng/ml). In addition, both *IL-6* and *IL-2* was established to play a crucial role as immunomodulators in acquisition of thermo tolerance during long-term heat stress exposure (Bharati et al., 2017). Further, the herbal supplements significantly reduced the heat stress induced increased expression of *IL-6* to the level similar to thermoneutral control group (C) reflecting the heat stress protection effects of the herbal products used in this study. Moreover the significant interaction between diet and heat treatment on the level of *IL-6* expression indicates the immuno modulatory role of the herbal supplements during heat stress exposure in Kenguri ewes.

Heat stress reduced the expression of *IL-18* in Kenguri ewes reflecting the sensitivity of this particular gene to temperature changes. It was established that the expression level of *IL-18* could be altered due to increased glucocorticoid secretion (Shini et al., 2010). The down-regulation effect on *IL-18* production in heat shock response is related to the suppression of the JNK/AP-1 signaling pathway (Wang et al., 2003). The *IL-18* is the pro-inflammatory cytokines actively involved in inflammatory response and the secretion of antibody under hyperthermia in animal models (Helwig and Leon, 2011). The down regulation of *IL-18* suggest that innate immunity was reduced under the condition of high temperature (Ma et al., 2019). The *IL-18* has a specific role in modulating the immune functions in animals and hence the reduction in *IL-18* can be a reflection of compromised immunity in heat stressed animals (Rashamol et al., 2019). Although the herbal supplement was used during heat stress in HSS group, still it could not restore the *IL-18* expression pattern in Kenguri ewes.

Heat stress significantly reduced the expression pattern of *IFN- β* in Kenguri ewes in this study. Similar findings of heat stress induced reduced *IFN- β* expression pattern was demonstrated in Salem Black goats (Madhusoodan et al., 2021). The *IFN- β* plays a crucial role in maintaining the innate immune response (Rashamol et al., 2019). Therefore, it could be inferred that the reduction in *IFN- β* in heat-stressed Kenguri ewes could reflect the compromised innate immunity in these animals. This suggests that the immune response is at stake even in extremely adapted indigenous Kenguri ewes during heat stress exposure. Further, the herbal supplements could not induce a positive impact on the expression pattern of *IFN- β* during heat stress exposure in Kenguri ewes.

Both *SOD* and *GPx* are classical antioxidant status reflecting genes in animals (Shi et al., 2020; Bai et al., 2020). Both these genes were significantly downregulated during heat stress exposure in Kenguri ewes in this study. It was shown that the activity of antioxidant enzyme could be depressed with the prolongation of the duration of heat stress, and this might be achieved by inhibiting antioxidant gene expression patterns such as *SOD* and *GPx* (Shi et al., 2020). In a similar study conducted in Salem Black goats Madhusoodan et al. (2020) demonstrated reduced *SOD* expression when subjecting the goats to chronic heat stress and attributed this to a sub-threshold level of the heat stress experienced by these goats. In a study conducted in broilers, Habashy et al. (2018) demonstrated reduced *GPx* expression during heat stress exposure and they attributed this to the increased cellular H_2O_2 as a result of heat stress which could result in the down-regulation of *GPx*. Surai (2006) observed that the increased *GPx* activity can be considered as an

additional protective mechanism under stress conditions. Thus, the downregulation of both *SOD* and *GPx* in this study in Kenguri ewes during heat stress exposure could reflect the highly compromised immune status in this breed. Although the heat stress reduced the *SOD* expression in HS group, still the herbal feed additives used in the HSS group could restore its expression in par with the control (C) group reflecting the immune protective effects of herbal supplement. However, the herbal supplements did not bring any positive impact in restoring the heat stress induced reduced expression patterns of *GPx* in Kenguri ewes.

Data from the study shows that heat stressed induced cortisol levels increased through middle of the experiment, then decreased towards the end of the experiment probably due to adaptation. However, samples were collected for gene expressions at the end of the experiment that animal partially adapted to the stress. Only one sample collected for gene expressions probably during adaptation period is a limitation of the study. Further, the animals used in the study were 15–23 kg (about 50 % difference between smallest and biggest animal) and thus this aspect also can be considered as a limitation of the study.

6. Conclusion

The study is a novel attempt to test the efficacy of herbal supplements on the immune response in Kenguri ewes during heat stress exposure. The higher plasma cortisol concentration recorded during heat stress exposure as compared to thermoneutral control clearly demonstrates the extremely severe heat stress experienced by Kenguri ewes in this study. Further, the herbal diet reduced the heat stress induced cortisol concentration. The impact of heat stress on the classical immunological genes in this study clearly demonstrated that the immune response is compromised even in extremely adapted Kenguri sheep. The increased plasma cortisol concentration reflected on the down regulation of all cytokines gene expression patterns except *IL-6* during heat stress exposure in Kenguri ewes. Thus, Cortisol, *IL-2*, *IL-6*, *IL-18*, *TNF- α* , *IFN- β* , *IFN- γ* , *SOD* and *GPx* can be considered biomarkers for reflecting the compromised immune response during heat stress exposure in Kenguri ewes. The herbal diet used in this study reduced the heat stress induced increased cortisol concentration and restored the heat stress induced reduced levels of *IL-2*, *TNF- α* , *IFN- γ* and *SOD* to normal range while it reduced the heat stress associated increased *IL-6* expression. This reflects the immune protective effects of herbal supplements used in this study. Thus, plasma cortisol, *IL-2*, *IL-6*, *TNF- α* , *IFN- γ* and *SOD* genes could serve as biomarkers for heat stress protection effects of herbal supplements used in this study.

CRedit authorship contribution statement

Rebez E. Binuni: Writing – original draft, Validation, Project administration, Methodology, Investigation, Data curation. **Veerasamy Sejian:** Writing – review & editing, Supervision, Project administration, Formal analysis, Conceptualization. **Dunshea F.R.:** Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Sahoo A.:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Nikhil Kumar Tej:** Writing – original draft, Validation, Methodology, Formal analysis. **Perumal S. Venkatesa:** Writing – original draft, Supervision, Methodology, Formal analysis. **Silpa M.V.:** Writing – original draft, Validation, Software, Resources, Methodology, Formal analysis. **Ninan J.:** Writing – review & editing, Validation, Supervision, Methodology. **Devaraj C.:** Writing – original draft, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that there is no any conflict of interest for this particular manuscript entitled “Effect of heat stress and herbal products supplementation on the immune response related gene

expression patterns in Kenguri ewes”.

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